

Research Report: Comparative Analysis of Chatbot Architectures

1. Introduction

The objective of this project is to develop and compare two distinct chatbot technologies: a rule-based system using **Artificial Intelligence Markup Language (AIML)** and a modern Large Language Model (LLM) system powered by **Ollama (Llama 3)**. This report analyzes their implementation, capabilities, and practical limitations.

2. Implementation Overview

2.1 Implementation A: Rule-Based (AIML)

The first chatbot utilizes **AIML**, an XML-based language for creating natural language software agents.

- **Mechanism:** It relies on pattern matching (`<pattern>`) and predefined responses (`<template>`).
- **Data Handling:** Uses tags like `<set>` and `<get>` to store specific variables (e.g., username, favorite color) in a symbolic memory.
- **Logic:** Hardcoded rules define the conversation flow.

2.2 Implementation B: LLM-Based (Ollama + Llama 3)

The second chatbot utilizes the **OpenAI Python library** as a client to interact with **Ollama**, running the **Llama 3** model locally.

- **Mechanism:** It uses a neural network-based transformer model to predict the next token based on a provided conversation history.
- **Data Handling:** Profiles are managed via a JSON file (`user_profile.json`), and context is maintained by passing a `conversation_history` list to the API.
- **Logic:** The behavior is guided by a "System Message" that defines the AI's persona and known user preferences.

3. Comparison of Capabilities

Feature	AIML Chatbot	Ollama (Llama 3) Chatbot
Response Generation	Fixed/Predefined.	Generative and dynamic.
Context Awareness	Limited to specific variables.	High (understands nuances in history).
Creativity	None. Only responds to known patterns.	High. Can brainstorm, code, and explain.
Error Handling	Uses a "Fallback" pattern for unknowns.	Can interpret typos and ambiguous intent.
Hardware Needs	Very low. Runs on any device.	Moderate to High (requires GPU/RAM).

4. Strengths and Limitations

4.1 AIML System

- **Strengths:** Extremely fast, predictable, and requires zero specialized hardware. It is excellent for "Command and Control" or FAQ systems where accuracy is vital.
- **Limitations:** It cannot handle phrases it wasn't programmed for. It lacks "common sense" and feels robotic.

4.2 Ollama (Llama 3) System

- **Strengths:** Highly conversational, multilingual, and capable of complex reasoning. It adapts to the user's tone and provides human-like interactions.
- **Limitations:** Risk of "hallucinations" (confident but false info). It is slower than rule-based systems and consumes more electricity/hardware resources.

5. Conclusion

While the **AIML** chatbot is reliable for specific, structured tasks, the **Ollama-based LLM** represents the current state-of-the-art in conversational AI. The LLM's ability to build a system message dynamically based on a JSON profile provides a sophisticated level of personalization that is difficult to achieve with static AIML rules. For general-purpose assistance, the LLM-based approach is superior, whereas AIML remains useful for lightweight, highly controlled environments.

